

Quantum Information

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Abstract

Extremely rough notes on quantum information. Maybe these will turn into something coherent, maybe not!

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1 POVM Measurements

1.1 Pure States and Projective Measurements

As typically introduced, the quantum mechanics of pure states $|\psi\rangle$ living in a Hilbert space $\mathcal{H}$ connects to experiments via projective measurements. Formally, a projective measurement is described by an indexed collection of projectors

$$\{\Pi_x : \mathcal{H} \rightarrow \mathcal{H} \mid x \in S\}, \quad S \subset \mathbb{R} \quad (1)$$

which add up¹ to the identity

$$\sum_{x \in S} \Pi_x = \mathbf{1} \quad (2)$$

so that they equivalently specify an internal direct sum decomposition

$$\mathcal{H} = \bigoplus_{x \in S} \mathcal{H}_x, \quad \mathcal{H}_x \equiv \Pi_x(\mathcal{H}). \quad (3)$$

A projective measurement then randomly projects any prepared pure state $|\psi\rangle \in \mathcal{H}$

$$|\psi\rangle \xrightarrow{\text{proj}} \frac{\Pi_x |\psi\rangle}{\|\Pi_x |\psi\rangle\|} = \frac{\Pi_x |\psi\rangle}{\sqrt{\langle\psi| \Pi_x |\psi\rangle}} \quad (4)$$

and the corresponding $x \in S$ is called the measurement outcome. Assuming WLOG that $|\psi\rangle$ is properly normalized $\langle\psi|\psi\rangle = 1$, the probability of any particular outcome is given by the Born rule

$$\text{Prob}_x = \|\Pi_x |\psi\rangle\|^2 = \langle\psi| \Pi_x |\psi\rangle. \quad (5)$$

If $S \subset \mathbb{R}$ is anything more analytically interesting than a finite set, this object should be interpreted with care as a probability measure. For the present discussion, it suffices to imagine that S is set.

¹Here “addition” is meant generously to include integration.

1.2 Entangled States and POVMs

The above is all well and good, but the following simple and natural thought experiment suggests a more general kind of process which does not quite take the form of a projective measurement, but which intuitively warrants being called a “measurement” of our system. Suppose we are interested in the physics of some Hilbert space $\mathcal{H}_A$, but that there exists a second space $\mathcal{H}_B$ and that we have the experimental power to perform general joint measurements. A general joint state

$$|\Psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (6)$$

is entangled, but for our purposes it will suffice to consider a pure state

$$|\Psi\rangle_{AB} = |\psi\rangle \otimes |\phi_{\text{aux}}\rangle \quad (7)$$

where we think of the second factor as an auxiliary state. Now suppose we perform a projective measurement on the joint state, described as above by a set of projectors

$$\{\Pi_{AB,x} \mid x \in S \subset \mathbb{R}\}. \quad (8)$$

The Born rule applied to the joint state gives the probability of outcome $x \in S$ as

$$\text{Prob}_x = (\langle\psi| \otimes \langle\phi_{\text{aux}}|) \Pi_{AB,x} (|\psi\rangle \otimes |\phi_{\text{aux}}\rangle). \quad (9)$$

If we think of this as a function of $|\psi\rangle$ at fixed $|\phi_{\text{aux}}\rangle$ and $\Pi_{AB,x}$, we can naturally rewrite the result as the diagonal matrix element of a quadratic form Q_x on $\mathcal{H}$ with matrix elements

$$Q_x : \mathcal{X}_A \times \mathcal{H}_A \rightarrow \mathbb{C}, \quad \langle\phi| Q_x |\chi\rangle \equiv (\langle\phi| \otimes \langle\phi_{\text{aux}}|) \Pi_{AB,x} (|\chi\rangle \otimes |\phi_{\text{aux}}\rangle). \quad (10)$$

The RHS is antilinear in the first argument and linear in the second, so this defines a bona fide quadratic form, and we can identify

$$\text{Prob}_x = (\langle\psi| \otimes \langle\phi_{\text{aux}}|) \Pi_{AB,x} (|\psi\rangle \otimes |\phi_{\text{aux}}\rangle) = \langle\psi| Q_x |\psi\rangle. \quad (11)$$

Recall that the polarization identity on quadratic forms

$$Q(u, v) = \frac{Q(u + v, u + v) - Q(u, u) - Q(v, v)}{2} \quad (12)$$

tells us a quadratic form’s entire action is fixed by its diagonal values, so knowing the value of $\langle\psi| Q_x |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}_A$ (or more weakly by linearity, just its values on unit-normalized such states) fixes Q_x uniquely.

The defining properties of $\Pi_{AX,b}$ as a projective measurement, plus judicious use of the polarization identity, imply that each Q_x is a Hermitian positive semi-definite operator²

$$Q_x = Q_x^\dagger, \quad Q_x \geq 0 \quad (13)$$

and that the collection taken together is a decomposition of the identity

$$\sum_{x \in S} Q_x = \mathbf{1}_A. \quad (14)$$

These two properties define a *positive operator-valued measurement* (POVM). The reader can convince themselves that any such collection $\{Q_x\}_{x \in S}$ with these two properties can be cooked up as we did here, by a suitable choice of auxiliary state $|\phi_{\text{aux}}\rangle \in \mathcal{H}_B$ and joint projective measurement $\{\Pi_{AB,x}\}_{x \in S}$.

²Here we are slightly redundant, as positive semi-definiteness is only defined for Hermitian operators to begin with.